Which fossil maximum ages for the origin of land plants can the sampled record test?

Internal research draft, Loop 2 sprint 2. Not for release; release and submission are Barry W. Brook's decision.

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Abstract

Molecular clocks for land plants rely on fossil maximum ages: claims that a lineage did not yet exist because its diagnostic fossils are absent from older rocks. Such a claim is testable only if the older rocks were searched in a way that would have found the fossils. We tested three maxima in current use under explicit sampling models:

- **M1, 515.5 Ma:** crown embryophytes, cryptospores;
- **M2, 459.1 Ma:** crown embryophytes, trilete spores;
- **M3, 431.2 Ma:** crown tracheophytes, polysporangiate bodies.

Only M2 is testable, and in two ways resting on different assumptions.

- **Murzuq Basin, Libya.** 28 Darriwilian–Hirnantian well cuttings without trilete spores, against 51 of 59 Llandovery cuttings with them. Together they exclude producers delivering triletes at Llandovery-reference levels for more than 3.5–8 Myr before 459.1 Ma, if older cuttings recovered spores at least a quarter as well as the reference. The conclusion depends on that transport ratio, which no source measures.
- **QSIM-801 core, Saudi Arabia.** No trilete spore was found in late Dapingian–early Darriwilian samples, among at least 99 examined cryptospore units. A trilete-to-cryptospore ratio above 0.031 in every sample is excluded at the 5% level, if triletes and cryptospores were recovered and recognised alike. This within-sample comparison needs no between-interval transport assumption.
- **What remains possible.** Neither record excludes trilete producers that were rare: below about a tenth of reference encounter in Murzuq, or at early-Rhuddanian rarity in QSIM-801.

M1 and M3 are not testable from the published record. M1 has no admitted design. Its occurrence route requires reading Cambrian Stage 4 cryptospore-like forms as embryophytic, an affinity that every describing source rejects. M3 has no searched-material denominator.

We give the observations that would make each maximum testable. We also give a general identification result: binary absence data cannot separate producer rarity from poor delivery or contaminated references.

1. Introduction

Fossil calibrations bound the ages of nodes in molecular-clock analyses. A minimum rests on a fossil that exists. A maximum rests on fossils that do not, in rocks old enough to have held them. The second kind of claim makes an implicit sampling argument: had the lineage existed, its fossils would have been preserved, recovered and reported. That argument is rarely written down and more rarely tested, although a maximum can shift a posterior divergence age as much as any minimum.

Land-plant clocks illustrate the problem sharply. Three maxima recur in recent analyses:

- **M1, 515.5 Ma, crown Embryophyta** (Morris et al. 2018; Su et al. 2021). It rests on Cambrian cryptospore-like palynomorphs.
- **M2, 459.1 Ma, crown Embryophyta** (Hooper et al. 2026). It rests on the oldest possible trilete spores, under an ancestral-state reading that the crown ancestor produced trilete monads.
- **M3, 431.2 Ma, crown Tracheophyta** (Hooper et al. 2026). It rests on Homerian *Cooksonia* cf. *pertoni* read as a total-group record.

The fossil record below each maximum is dominated by palynological samples from marine and marginal-marine rocks, gathered for biostratigraphy rather than to search for land plants. Whether those samples can carry a maximum is an empirical question about their observation process.

We asked, for each maximum: is there a set of older sampled units, reported completely enough to count as absences, together with a comparable reference in which the target fossils are reliably recovered? If so, what origin ages and abundances does the record exclude, under which explicit assumptions? If not, which specific observation is missing? We treat a maximum as a claim about encounter: how often a producer's diagnostic remains reach and are recovered from sampled rock. Encounter is distinct from origin. A finite bound on encounter does not by itself date a node.

2. Methods

2.1 Events, readings and outcome coding

Events. Each maximum maps to an observable event:

- **E1:** trilete spores;
- **E2:** cryptospores accepted as embryophytic;
- **E3:** polysporangiate bodies.

Readings. Disputed identifications are carried as named readings rather than resolved by fiat.

- **main:** the default identification of each disputed record;
- **inclusive:** wider acceptance of disputed forms as the event;
- **outlier:** isolated records treated separately.

Coding rules.

- **Same rule in both intervals.** Within a frame (a study's complete sample inventory), older and reference units are coded by the same rule.
- **Detections and zeros.** A detection maps to one. An explicit non-detection maps to zero only within its verified scope. A not-reported outcome becomes zero only after a full-category reporting audit.

- **Unresolved records never become zeros.** This includes disputed records, unknown outcomes and missing sample links.
- **E2 identifications** follow a two-axis rule (Supplement S2):
  - the morphology and producer evidence the source reports;
  - the embryophyte affinity the named reading accepts.

  A cryptospore label alone does not establish E2.

2.2 The encounter model and its qualification (Murzuq)

Unit model. Each productive sampled unit has a detection probability $1 - \exp(-r\,u\,x)$.

- r is the reference encounter rate;
- u ~ Gamma(k, k) is a well effect shared by all units of a well (k = 2);
- x is the unit's relative encounter intensity.

For older units, $x = \rho f$. Here f is the producers' intensity relative to the reference, and ρ is the older-to-reference delivery-and-recovery ratio. ρ is a declared scenario (1, 0.5, 0.25, 0.1), never estimated.

Quantities reported.

- **Origin comparisons:** a constant intensity (H0c), exposure-weighted (H0, H0m) and flexible (H1) intensity bounds at 459.1 Ma;
- **Window comparisons:** the oldest age to which reference-level encounter (f = 1) could have persisted before 459.1 Ma without contradicting the older zeros.

Screen and qualification.

- **Potential screen.** A frame is analysed only if it passes a likelihood-ratio screen against 5.991, so that it could in principle reject a reference-level history.
- **Qualification.** Each released quantity was qualified by simulating the full eligibility, screen and estimation path on a declared finite grid of histories, against a 0.93 coverage floor (nominal 0.95; calibration target 0.955).
- **Abstentions** count as covering.
- **Reporting.** We report these as nominal 95% compatibility bounds. Coverage conditional on a non-abstaining result was not established (Supplement S1).

2.3 Identification and robustness without counts (sprint 1)

The Murzuq outcomes are binary marks on range charts. Its per-sample palynofacies percentages count 200 organic particles on separate preparations and cannot be paired with the trilete marks. We therefore extended the binary model rather than fitting counts.

Extended model.

- A fraction g of reference encounter is indigenous; the remainder is contamination (for cuttings, mainly caving).
- Contamination reaches older units at relative intensity κ.
- An older unit's zero then has probability $\exp[-r\,u\,\kappa(1-g)]\cdot\{(1-q) + q\cdot\exp(-r\,u\,\rho\,g\,f)\}$, where q is the probability that the unit falls in the target interval.

Grid and comparisons.

- **Declared before computation:** a 5,760-row grid over ρ, g, κ, Hawaz age allocation, four chronology scenarios, four targets and three intensities.
- **Shared-age scenarios:** fixed, depth-ordered positions at the 10%, 50% and 90% points of each unit's age range.
- **Comparisons** use the likelihood-ratio convention −2 log 0.05 without calibration.
- **Status:** they are unqualified sensitivities, not confidence statements.

2.4 A within-sample share bound (QSIM-801)

The observations. Vecoli et al. (2017) examined 20 core samples from the QSIM-801 well. Their Table 1 lists the cryptospore species seen in each sample. They state that "only cryptospores and no trilete spores have been observed" (p.330). One examination per sample produced both observations.

The lower bound. Each occurrence mark is at least one examined cryptospore unit. The 99 marks across 17 positive samples therefore lower-bound the land-spore units examined.

The bound. Let θ_i be sample i's trilete:cryptospore ratio among units reaching the slides. Let h be the probability that a trilete unit is recovered and recorded, relative to a cryptospore unit.

- **Model.** Under independent Poisson sampling, conditioning on each sample's land-spore total N_i gives a binomial trilete count.
- **Inequality.** The probability of no triletes is $\prod(1 + h\theta_i)^{(-N_i)} \le (1 + h\theta)^{(-M)}$, where θ is a floor holding in every sample and $M = 99 \le \Sigma N_i$.
- **Test.** Rejecting θ when $(1 + h\theta)^{(-99)} \le 0.05$ is a valid level-0.05 test of that persistent floor.

What it is not. It is not a bound on a mean ratio: heterogeneity between samples makes zeros more likely.

Assumptions.

- **Parity scenarios:** h = 1, 0.5, 0.25 or 0.1. Within-sample clustering enters as a design effect that divides M (1, 2, 4).
- **Known-answer check.** Simulation verified the test's level when the true examined count far exceeds the marks, and in a stress regime where marks nearly equal counts (Supplement S3).

We did not use Le Hérissé et al.'s (2017) palynomorph percentages. Their preparations and counts cannot be linked to Vecoli et al.'s examination (Codex, sprint-1 gate).

2.5 Source adjudication for M1 and M3

M1. For the M1 occurrence route, we applied the E2 rule to every held source describing the oldest cryptospore-like forms. Each source's morphology and stated affinity were recorded separately, with page locators.

M3. For M3, we assessed whether any held source records searched rock or beds that yielded no polysporangiate fossils, together with their effort.

3. Results

3.1 M2 in the Murzuq Basin

The frame. The admitted cutting frame holds 28 older units (Darriwilian–Hirnantian), none with a trilete report, and 59 Llandovery reference units, 51 with triletes (Abuhmida 2013).

Origin comparison. At ρ = 1, the constant-intensity bound at 459.1 Ma is H0c f* = 0.0955. The exposure-weighted H0 bound is f* = 0.189.

Window endpoints. Reference-level encounter (f = 1) is excluded for longer than a short window before 459.1 Ma.

- **Endpoints by ρ:** the window ends at 462.6, 464.1 and 467.1 Ma at ρ = 1, 0.5 and 0.25. It is unbounded at ρ = 0.1.
- **Low encounter.** Windows at f = 0.1 and 0.01 are unbounded at every ρ: low encounter is compatible with any origin age.
- **Qualification.** All fourteen qualified combinations pass the finite-grid contract: thirteen Murzuq design/ρ pairs and one core comparison without an older interval.

Boundary case. At the edge of the real 462.6 Ma claim, 20 of 1,000 simulated passing records gave a bound excluding the truth. Those 20 are a third of the 60 records that gave any window claim.

Robustness (Table 1; Figure 2).

Table 1: What changes the Murzuq answer. Values from `models/f3/loop2/out/`.

Assumption	Effect at ρ = 1	Status
Leave one well out	H0 bound 0.178–0.365; window endpoint 462.6–468.1 Ma (omitting H2 gives 468.1)	unqualified sensitivity
Reference contamination, no transfer to older units (κ = 0), independent uniform ages	462.6 Ma window flips when indigenous fraction $g < 0.956$; 467.1 Ma window at $g < 0.236$; origin comparison at $g < 0.091$	unqualified sensitivity
Shared midpoint ages	459.1–462.6 Ma has no sampled exposure	identification limit
Rarity against delivery	the likelihood depends on biology and transport only through $a = \rho g f$ and $b = \kappa(1 - g)$	identification limit

What the robustness work shows.

- **The short window is fragile.** Under 5% contamination of reference positives defeats it, and shared midpoint ages remove its exposure altogether.
- **The wider statements are robust.** The origin comparison and the 467.1 Ma window survive substantial contamination.
- **The general limit.** Producer rarity, poor delivery and contaminated references are observationally equivalent along $a = \rho g f$. Binary absences cannot bound indigenous production without an assumption on delivery and reference purity.

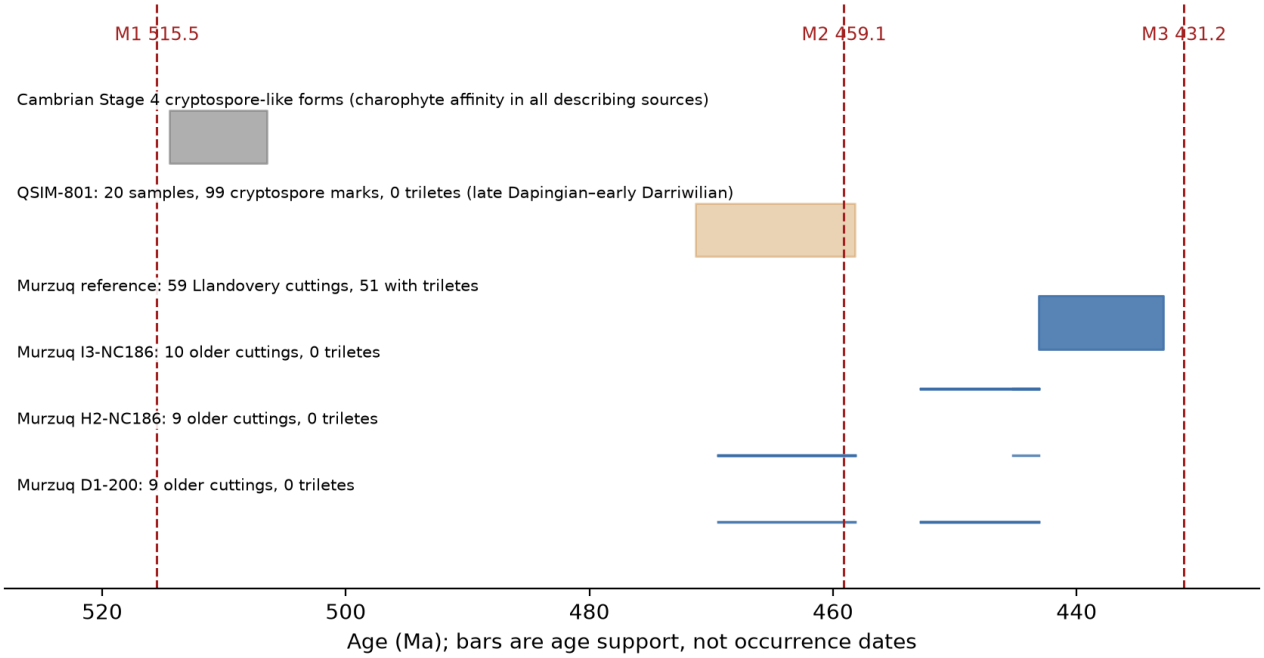


Figure 1: Sampled support relative to the three maxima. Bars are age support, not occurrence dates.

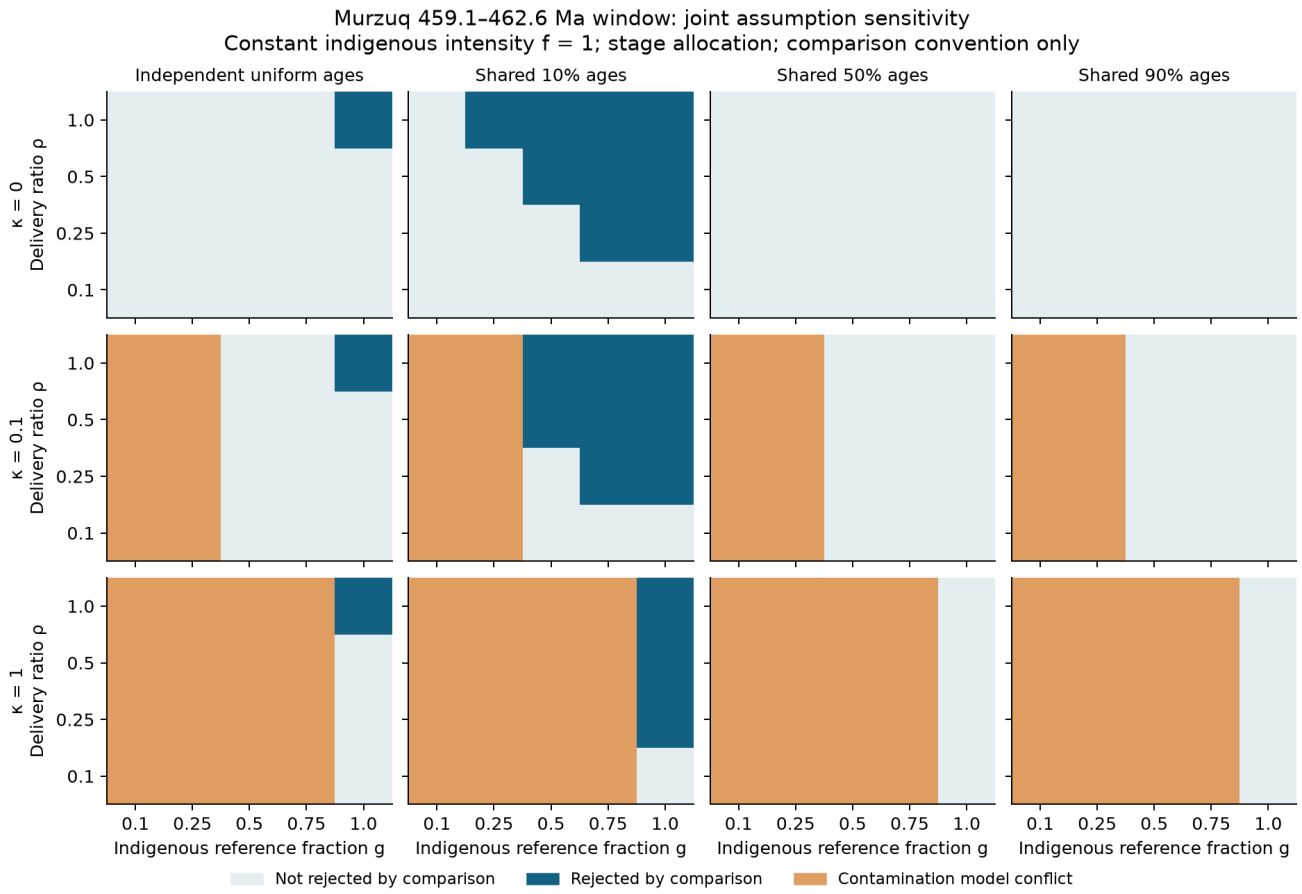


Figure 2: Sensitivity of the short Murzuq window to reference contamination and chronology (sprint 1).

3.2 M2 in central Saudi Arabia (QSIM-801)

The bound. At h = 1, a persistent trilete:cryptospore ratio θ > 0.0307 is excluded: the recovered trilete share is below 3.0% of land-spore units (Table 2; Figure 3). The bound scales as 1/h. At h = 0.1, triletes being recovered or recognised ten times less readily than cryptospores, it is θ < 0.31.

Table 2: Upper 95% bound on a persistent trilete:cryptospore ratio θ, QSIM-801.

h	Design effect 1	2	4
1	0.031	0.062	0.129
0.5	0.061	0.125	0.257
0.25	0.123	0.250	0.515
0.1	0.307	0.624	1.287

The reference comparison is qualitative. No held source gives counted trilete and cryptospore tallies for a Katian–Llandovery reference. We scanned 346 held texts. The published descriptions are:

- **Early Rhuddanian:** a single trilete among “very rare” cryptospores in one Murzuq assemblage (Abuhmida 2013, p.382);
- **Another Ashgillian–Rhuddanian-correlated zone:** triletes “moderately abundant” (Abuhmida 2013, p.383);
- **Mid-Telychian onward:** triletes dominate assemblages (Gray et al. 1992, as summarised in Abuhmida 2013, p.350).

What it excludes, and what it doesn’t.

- **Excluded:** a trilete-dominated land-spore rain ($\theta \geq 1$) in the late Dapingian–early Darriwilian, under every recovery scenario down to $h = 0.031$.
- **Not excluded:** early-Llandovery rarity.

Relation to Murzuq. The Saudi result shares the Murzuq conclusion: abundant trilete producers are excluded, rare ones are not. It rests on a different assumption, within-sample recovery parity rather than between-interval transport.

What it is not.

- **Not a replication:** the estimands differ, and no counted reference makes them comparable.
- **Not a qualified primary result:** it is an empirical bound under declared parity.

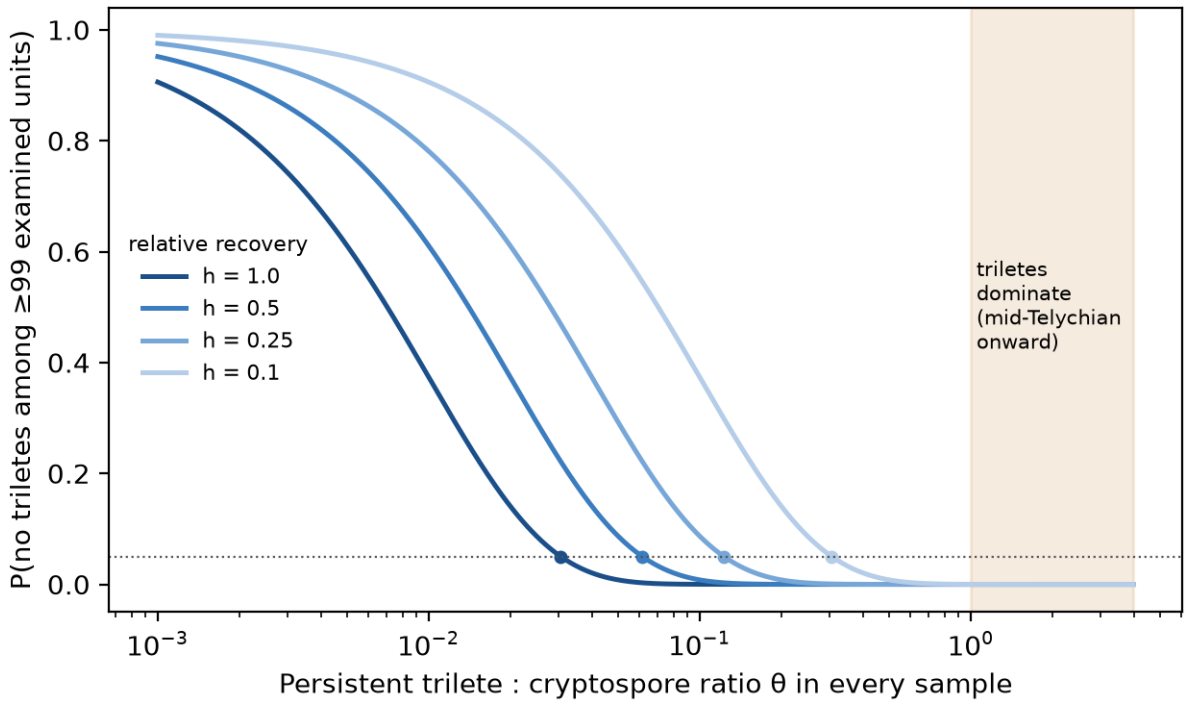


Figure 3: Probability of no triletes among at least 99 examined cryptospore units, against a persistent trilete:cryptospore ratio, for four recovery-parity scenarios. Points mark the 5% bound.

3.3 M1: no admitted design, and an occurrence route the describing sources reject

No admitted design. No assessed frame supplies an older-zero/reference design for E2. The one secure older zero (Borenshult 71.1 m) belongs to a frame that fails the potential screen (LR 2.21). Disputed Cambrian–Floian forms cannot become zeros.

The occurrence route. The alternative is to read the Cambrian Stage 4 forms (514.5–506.5 Ma) as E2 itself. Under the E2 rule this requires embryophyte affinity, which the held describing sources reject:

- **Strother (2016)** places the Tennessee taxa in Charophyta (pp.28, 32);
- **Taylor & Strother (2024)** interpret the Rome Formation packets, coeval with the oldest North Chinese forms, as "pre-embryophytic spore types" (p.90). They read them as the products of terrestrial charophytic algae, "likely not produced in a sporangium" (p.103);
- **Wang et al. (2022)** judge the Shandong Stage 4 forms "unlikely directly related to embryophytes" (p.6).

Verdict. The inclusive reading therefore rests on morphology alone. It also comes close to assuming what M1 tests: the clock's own prediction of embryophytes by about 515 Ma is the main reason to read these forms as embryophytic. M1 remains an absence question. Testing it needs productive, completely reported material older than 515.5 Ma; none is held.

3.4 M3: genuine negative evidence without a denominator

Edwards & Feehan (1980) report fossil-free interlaminated mudstone beside *Cooksonia*-bearing siltstone in Ireland. This is real negative evidence, but it has no n.

The Czech material comprises about fifty historical axes and eleven new sterile fragments. These are selected plant specimens, not searched beds (Libertín et al. 2018).

Databases. Of the 3,217 GBDB sections, 3,154 have no recorded core plants. Those are database omissions, not E3 non-detections.

What would admit a design. A field or collection ledger of searched beds, with effort and preservation, together with a comparable reference.

4. Discussion

What is testable

Of three maxima, one can be tested. It is testable under two assumptions that differ in kind, and the two agree. Trilete producers that were abundant and delivered spores as the Llandovery flora did are excluded:

- from the Murzuq shelf for more than about 3.5–8 Myr before 459.1 Ma;
- from the central Saudi land-spore rain of the late Dapingian–early Darriwilian, older than 458.2 Ma.

Producers that were rare, or poorly delivered relative to the reference, are compatible with any origin age in both records. The record is just as empty immediately after 459.1 Ma: the Murzuq H0 read-out bounds post-maximum encounter below about 0.19 of the reference. The absence argues against an old, abundant crown flora. It does not date the crown.

Why rarity and delivery cannot be separated with absences

The sprint-1 identification result is general. In any binary design with a shared well effect, the older zeros constrain only the product of delivery, indigenous reference fraction and producer intensity. A contaminant fraction enters separately. So a maximum built on absence is only as strong as an independent constraint on delivery and reference purity.

The within-sample design changes which assumption carries the weight. Cryptospores in the same slides show that land-derived spores did arrive and were recovered, so the ratio test needs only recovery parity between two spore types of similar size. Taken alone, its reference comparison is qualitative.

The single most valuable new observation is a counted Katian–Llandovery assemblage with separate trilete and cryptospore tallies, from a comparable facies. It would turn the Saudi bound into a quantitative comparison with the reference flora.

What the clock analyses should take from this

- **M2.** As a soft maximum, M2 is defensible against abundant, well-delivered trilete producers. It is not defensible against a rare early lineage, and its 462.6 Ma window should not be read as a hard edge.
- **M1.** M1 cannot be tested by absence from the published record. Its occurrence basis, if any, rests on an affinity its describers reject.
- **M3.** M3 has no denominator.

None of this argues that the maxima are wrong. It shows which parts of each are evidence and which are assumption. The specific missing observations are:

- **M1:** complete rosters of productive pre-515.5 Ma samples;
- **M3:** searched-bed ledgers for the Irish and Loděnice sections;
- **M2:** counted reference shares and matched contamination diagnostics.

Limitations

- **Single studies.** Each M2 frame comes from one study: one author or laboratory, and one chart system or table.
- **ρ and h are declared, not measured.**
- **Contamination.** The contamination model is a scenario model, not an estimate of caving.
- **Qualification is finite-grid.**
  - Coverage given a claim, simultaneous coverage across curves and validity beyond the tested histories are not established.

  - The QSIM-801 bound is not qualified by the Loop 1 contract. It is an exact test whose level was checked by simulation.

- **Age support versus age.** Age support is used where occurrence dates are unavailable.

Roles of AI systems and humans

- **Barry W. Brook** set the programme and its principles, authorised each stage, and reviews the final work. This draft has not yet had that review.
- **Claude Opus 5.5** (Anthropic, `claude-opus-5-5`, via Claude Code):
  - drafted the F3 framing, the cards and the Loop 1 close;
  - in Loop 1, froze the M2 designs and wrote the engine extensions, adapter, screen and actual-design qualifier;
  - wrote Loop 2 plan v2;
  - reviewed sprint 1 and independently reproduced its break-even thresholds;
  - derived and implemented the QSIM-801 bound;
  - adjudicated the M1 affinity sources;
  - wrote this manuscript.
- **OpenAI Codex** (Codex desktop; GPT-6, exact variant not exposed):
  - proposed and executed Loop 2 sprint 1 (observation audit, identification result, sensitivity grids, count-share method);
  - ran Loop 1's M1/M3 source admission, then counter-reviewed and requalified the M2 designs;
  - corrected the coverage statement and the cryptospore definition during reconciliation.
- **Claude Sonnet 5.5 subagents** and **Codex delegates** performed bounded extraction, simulation and audit tasks, each verified by the parent agent.

The full record, including the human–AI correspondence, is in the repository (`docs/charter.md`, `docs/decisions.md`, `docs/audits/`).

Data and code

All derived tables, code, qualification receipts and figure scripts are in the repository. Copyrighted source texts are held locally only. `manuscript/data-manifest.csv` separates distributable derivatives from private-reproduction inputs. `manuscript/claims.csv` links every headline number to its evidence and status.

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Supplement

- **S1. Coverage statement and qualification receipts.** `docs/loop1-close.md`; `models/f3/out/qual/`.
- **S2. E2 rule.** `data/units/e2-unit-rule.md` (v1).
- **S3. QSIM-801 extraction, bound and known-answer check.** `data/units/loop2/replication/`; `models/f3/loop2/replication/`.
- **S4. Sprint-1 observation audit, identification proof and sensitivity grids.** `docs/loop2-sprint1-results.m` `models/f3/loop2/`.
- **S5. M1 affinity adjudication and drafted data requests.** `data/units/loop2/m1-m3/`.
- **S6. Claim-to-evidence table** (`claims.csv`) and data manifest (`data-manifest.csv`).